

Uma Shankar Kommireddy

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Professional Summary

AI/ML & Full Stack Engineer with 2 years and 2 months of experience building LLM-powered applications, RAG pipelines, backend microservices, and full-stack features using Python, FastAPI, Node.js, and TypeScript. Hands-on experience with LLM integration, semantic retrieval, prompt engineering, vector search, and privacy-preserving AI, alongside React frontend development. Strong foundation in REST/GraphQL APIs, PostgreSQL, Redis, Docker, Kubernetes, and distributed systems, with experience taking features from database and API design through AI processing and user-facing interfaces.

Technical Skills

Languages: Python, TypeScript, SQL, C++

AI & Applied AI: LLM Integration, AI Agents, Agentic Workflows, RAG, MCP, LangChain, LangGraph, Ollama

Backend APIs: MolecularJs, FastAPI, Django, Node.js, REST APIs, GraphQL, Microservices, Server-Sent Events

Frontend: React, TypeScript, JavaScript, HTML, CSS, Material UI

Databases & Retrieval: PostgreSQL, MySQL, Redis, pgvector

Cloud & Production Systems: Docker, AWS (Basics), Git, GitHub Actions, GitLab CI/CD, Kafka

Security & Data Privacy: PII Detection, Data Anonymization, Data Privacy, Presidio, Semgrep, OWASP ZAP

Experience

Guidizy

Jul 2025 – Aug 2026

Software Engineer (Full Stack & AI)

Remote

- Designed and shipped reusable **LLM/AI integrations** using RAG and Model Context Protocol (MCP) for agentic workflow automation across 10+ backend microservices.
- Replaced manual FAQ-node creation with an AI-driven pipeline that automatically generates FAQ nodes from customer-uploaded knowledge-base PDFs, reducing manual effort by **~99%**.
- Built asynchronous execution pipelines using **Redis, Server-Sent Events, Node.js, TypeScript, and PostgreSQL** to support real-time workflow execution.
- Implemented **PII detection, data anonymization, Semgrep, and OWASP ZAP** to improve privacy and security across AI processing and CI pipelines.

QualityKiosk Technologies

Apr 2024 – Jun 2025

Digital Quality Engineer

Mumbai

- Validated REST APIs and backend data integrity using **SQL verification and regression testing** for a banking wealth-management platform, resolving **70+ Jira tickets** while collaborating with engineering teams to triage defects.

Projects

AI Financial Assistant (Github)

Python • AI and LLM • FastAPI • PostgreSQL • Redis

- Designed an AI financial assistant architecture targeting **1M+ banking Transactions** and **1000 users**, combining LLM orchestration, deterministic financial data retrieval, semantic search, and personalized AI memory.
- Built a **hybrid AI retrieval architecture** that routes exact financial queries to PostgreSQL while using pgvector-based semantic retrieval for contextual queries, reducing hallucination risk by grounding responses in verified and user data.
- Designed a **3-layer context system** combining short-term conversation history, verified financial data, and long-term user memory, with individual memory buckets capped at 50 items to prevent unbounded context growth.
- Implemented a **security-first request pipeline** with PII anonymization, JWT-based authentication, token-bucket rate limiting, user-scoped retrieval, and SSE streaming, protecting sensitive banking data across the AI processing lifecycle.

RAWJD – AI Resume Intelligence Platform (Github)

LLM/RAG • React • Express • Redis • TypeScript

- Built an end-to-end **LLM-powered application** using semantic retrieval, evidence ranking, context caching, and prompt optimization to ground AI outputs in resume and JD evidence.
- Reduced AI prompt size by **65–75%** and token consumption by **60–70%**; implemented privacy-first PII tokenization with encrypted Redis mappings and automatic 6-hour expiration.
- Built the **React frontend** and TypeScript backend with REST APIs, GraphQL, SSE, and Redis session management.

Achievements

- Finalist, Bessemer Tech Catalyst Hackathon** – Selected among the top 300 finalists from 10,000+ registrants (Top 3%).
- HackerRank** – 5-Star rating in Problem Solving and Python.

Education

Sagi Rama Krishnam Raju Engineering College

2021 – 2024

Bachelor of Technology in Computer Science